

1. Geometry of Two-Input Perceptron:

A. two-input perceptron can visualize by using (x_1, x_2, z) as 3 parameters of (x, y, z) in 3D space, the perceptron will form a "plane" in 3D space.

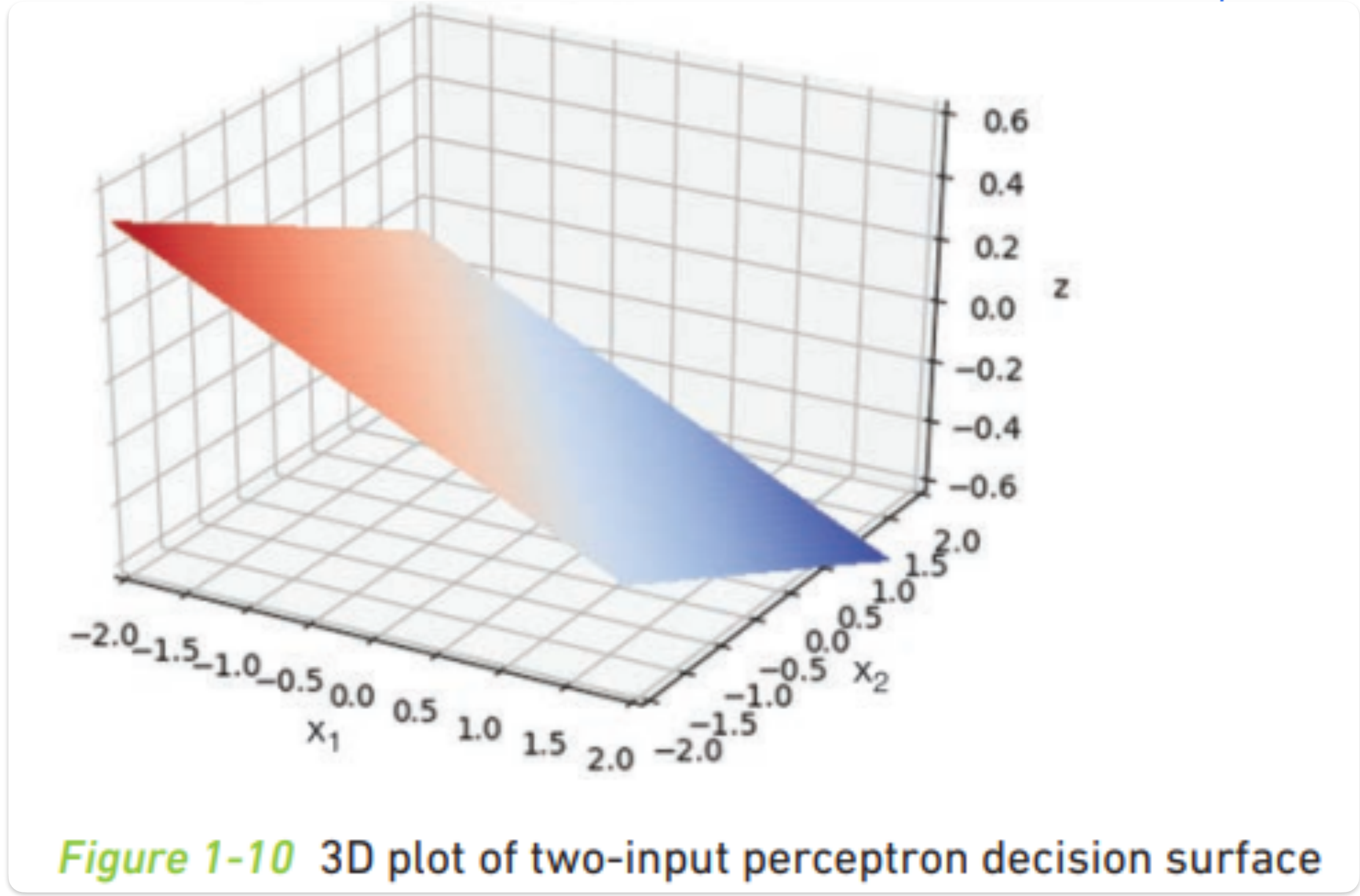


Figure 1-10 3D plot of two-input perceptron decision surface

B. looking at this 3D plane from above and draw the intersection line with $z=0$, we can get 2D straight line in 2D space.

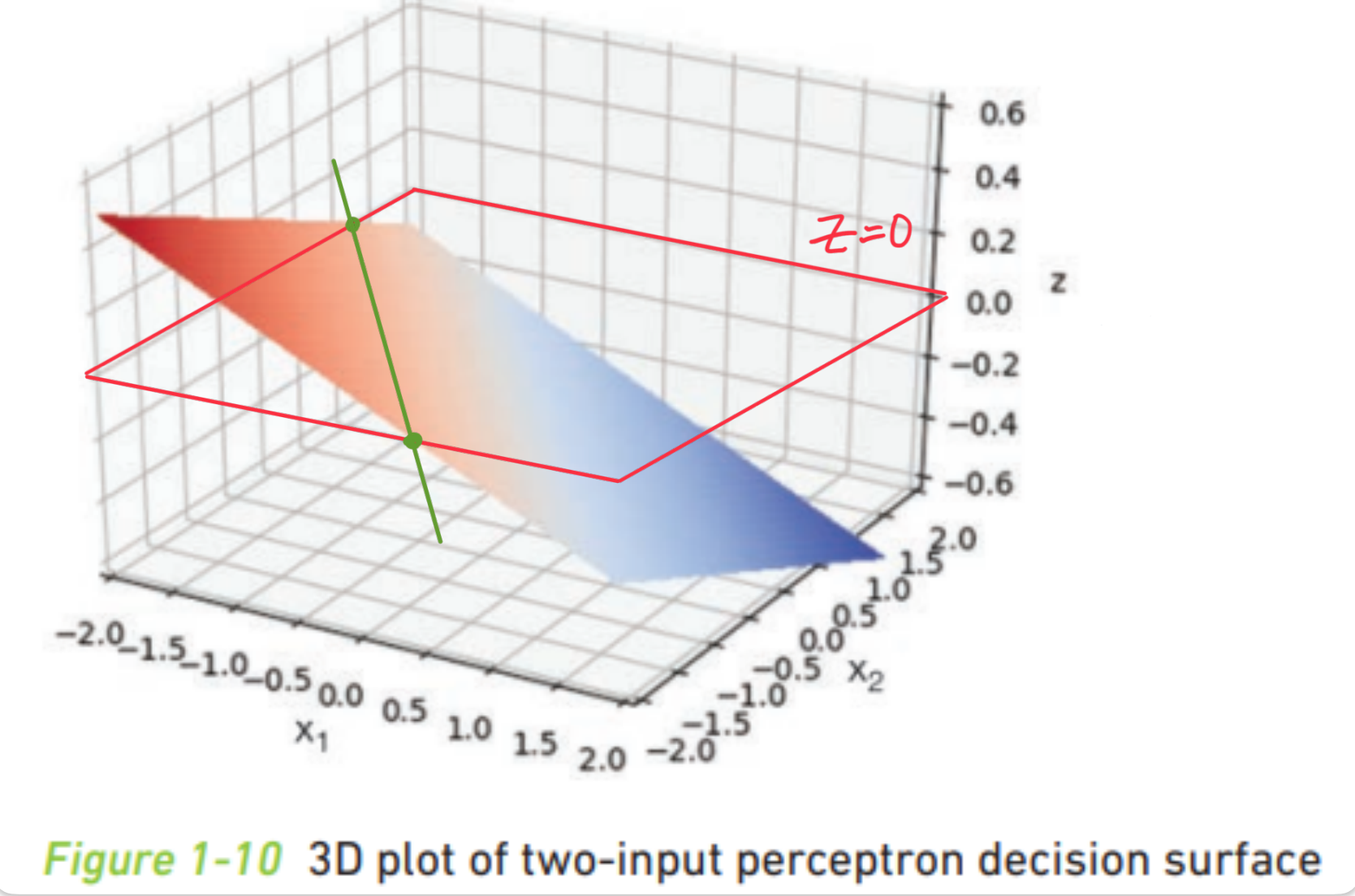


Figure 1-10 3D plot of two-input perceptron decision surface

3D space

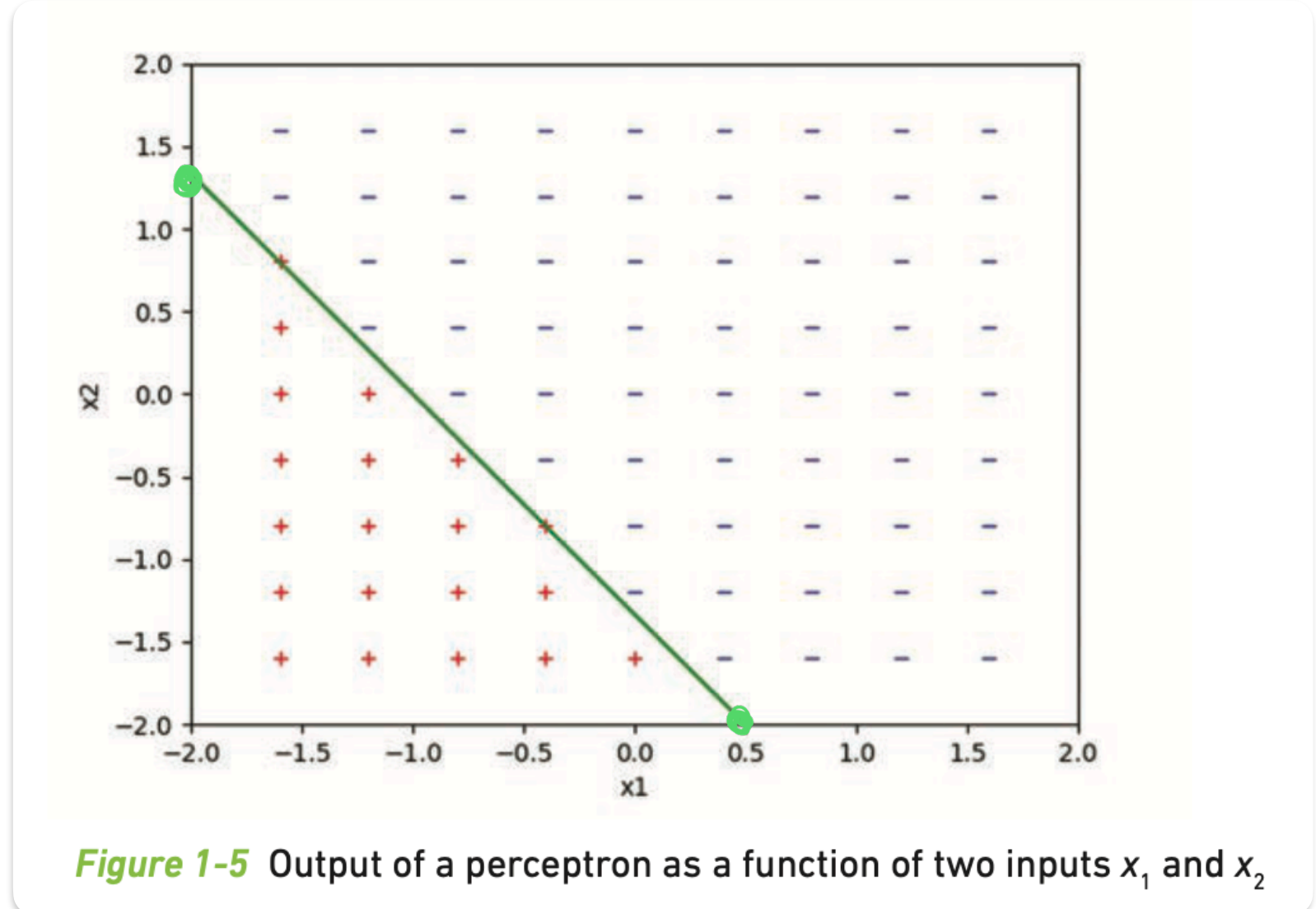
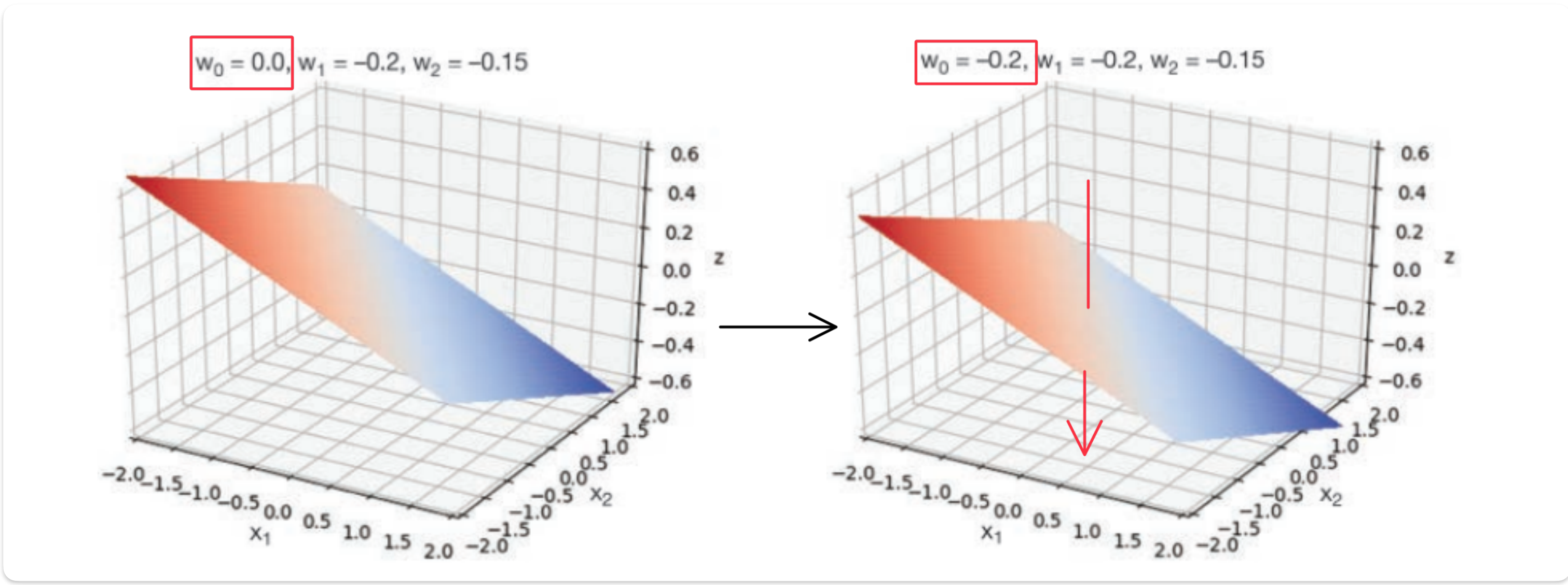


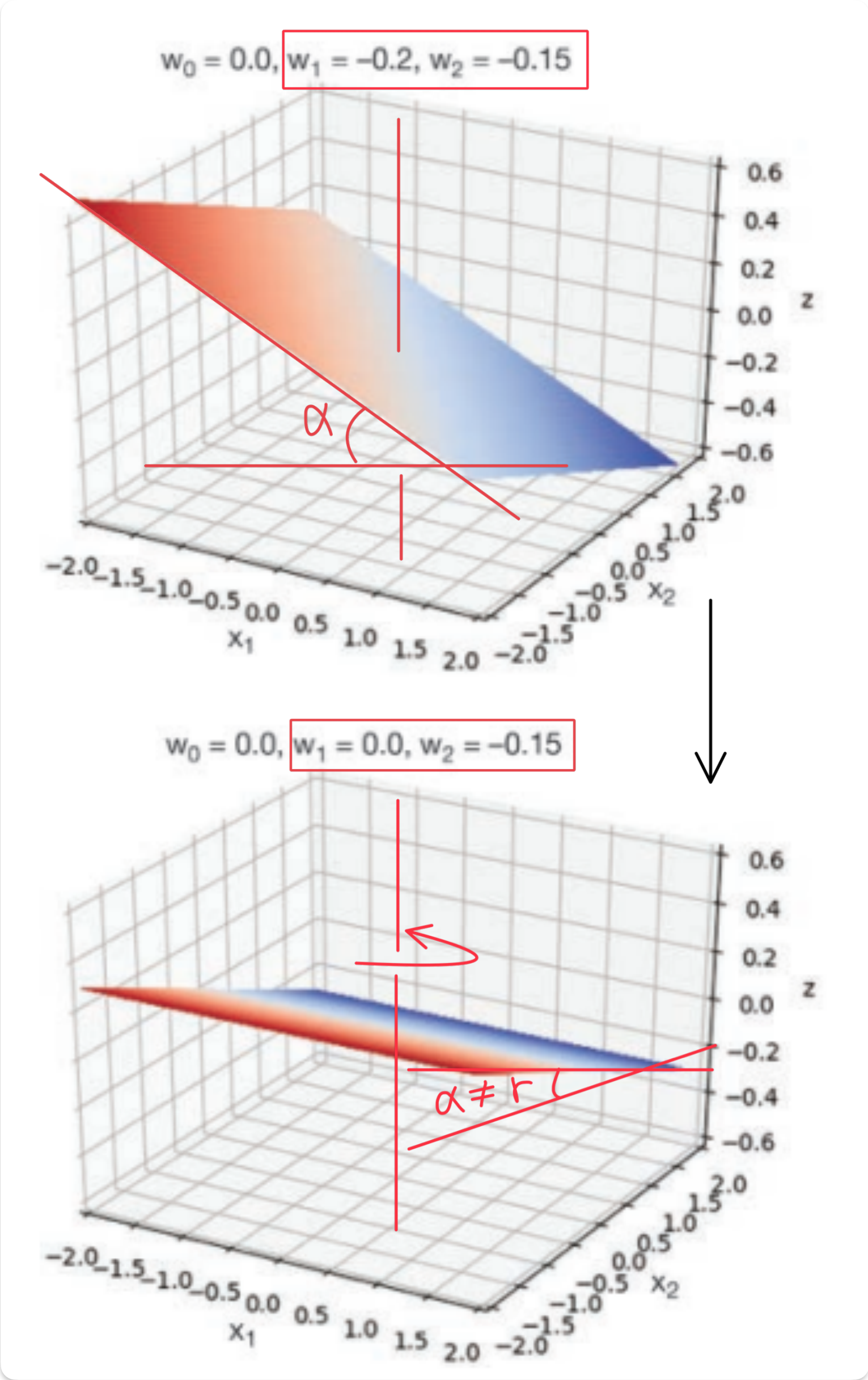
Figure 1-5 Output of a perceptron as a function of two inputs x_1 and x_2

2D space

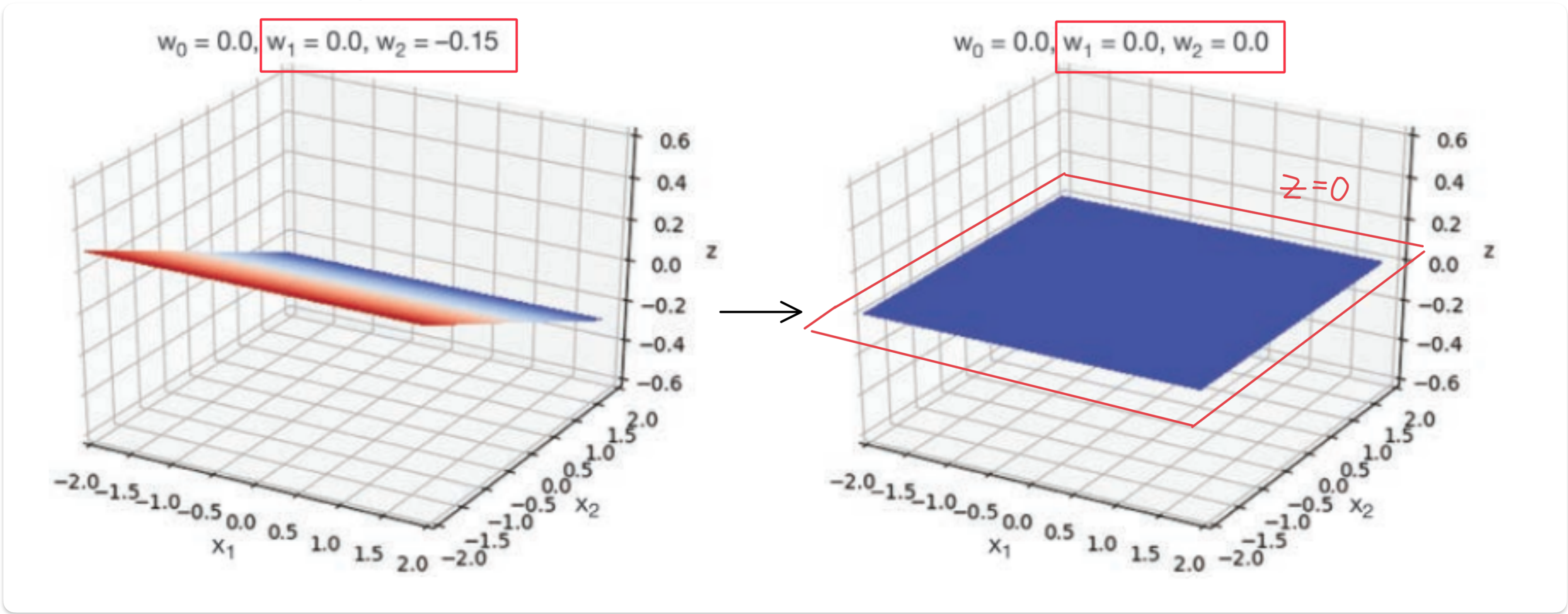
C. bias weight (w_0) of two-input perceptron can determine the plane to move up or down in 3D space.



D. two other weights (w_1, w_2) of two-input perceptron can determine the plane to slope and change its direction in 3D space.



E. if two other weights (w_1, w_2) of two-input perceptron are all 0, the plane will parallel to $z=0$.



2. Concept of Hyperplane:

if the perceptron is linearly seperable, it can distinguish between 2 classes.
for a two-input perceptron means 2 classes. can be separated by a straight line with x_1, x_2 in 2D space.
for a three-input perceptron means 2 classes. can be separated by a flat plane with x_1, x_2, x_3 in 3D space.
for a n-input perceptron means 2 classes. can be separated by a n-1 dimension hyperplane with $x_1, \dots, x_n$ in nD space.

3. About vector $\vec{w}$ ($\vec{w}$ & $\vec{w}^T$ are the same vector in different format) and vector $\vec{x}$:

in a perceptron, $z = \vec{w}^T \cdot \vec{x} \equiv \vec{w} \cdot \vec{x}$,
 $\vec{w} \cdot \vec{x} = |\vec{w}| \cdot |\vec{x}| \cdot \cos\theta$, θ is the angle between $\vec{w}$ and $\vec{x}$, so θ can determine the final output ($y = f(z)$) of perceptron,
 $\theta > 90^\circ \rightarrow \cos\theta < 0 \rightarrow z < 0 \rightarrow y = -1$
 $\theta \leq 90^\circ \rightarrow \cos\theta \geq 0 \rightarrow z \geq 0 \rightarrow y = +1$